

WHY SOME THUNDERSTORMS HAVE EXCESSIVE AMOUNT OF LIGHTNING

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A thunderstorm is defined as convection that has at least one stroke of lightning that produces audible thunder. You may have noticed that some thunderstorms have much more lightning and thunder than others. Why is this?

There are three factors that contribute to a storm having an exceptionally large amount of lightning and thunder. Each of these will be discussed individually.

HIGH INSTABILITY RELEASE: High instability is a condition in which the ambient tropospheric temperature decreases rapidly with height, especially in the top lower to mid-levels of the troposphere. When instability is high, thunderstorm updrafts will be more intense. The stronger the thunderstorm updraft, the deeper the thunderstorm column will be. As air rises in a thunderstorm it cools. When the storm height is very high, the top of the thunderstorm will cool to very cold temperatures. This intense cooling glaciates the top of the storm and this can be seen as the thunderstorm anvil. The glaciation process produces a charge differential in the storm cloud. In cases where very rapid and intense glaciation occurs (very high CAPE), lightning and thunder will be generated to a significant degrees. All thunderstorms have ice in the upper portions of the storm. How fast the ice develops, the depth of the icy portion of the storm and how quickly the precipitation moves within the cloud are important to the lightning process. There is still much research that needs to be done to fully understand this process.

HIGH MOISTURE CONTENT: Operational meteorologists determine the potential moisture a thunderstorm will have by examining lower tropospheric dewpoint and the precipitable water (PW) in the troposphere. Since thunderstorm updrafts often begin in the lower portions of the troposphere, it is most important to examine moisture there. Dewpoints of 60 F or greater in the lower troposphere will bring significant moisture into a storm. Dewpoints of 70 F or higher in the lower troposphere are not uncommon in the warm season maritime tropical environment. Low level moisture helps in that it increases instability. This in turn helps lead to a stronger storm updraft. An increase of moisture also means more ice can be produced when the moisture begins to glaciates in the updraft. Charge differential builds up more significantly as the mass of ice and water in the thunderstorm cloud increase.

WIND SHEAR: Wind shear is wind speed changing significantly with height and/or wind direction changing significantly with height. Wind shear enables a thunderstorm to last for a longer period of time since it helps displace the updraft from the downdraft. These thunderstorms are often in the form of multi-cell storms or supercell storms. Wind shear also increases turbulence within the thunderstorm. This violent mixing of precipitation in the air could help enhance charge separation in a storm.

Typical profile for highly active lightning storm: CAPE: 3,000 J/kg or greater, low level dewpoints greater than 65 F, PW 2.00 inches or greater, change in wind speed and direction with height.